How signers signal event boundaries

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Research Question

Do **theoretical** phonological and motion ASL features **quantitatively** signify telicity? How do they **interact**?

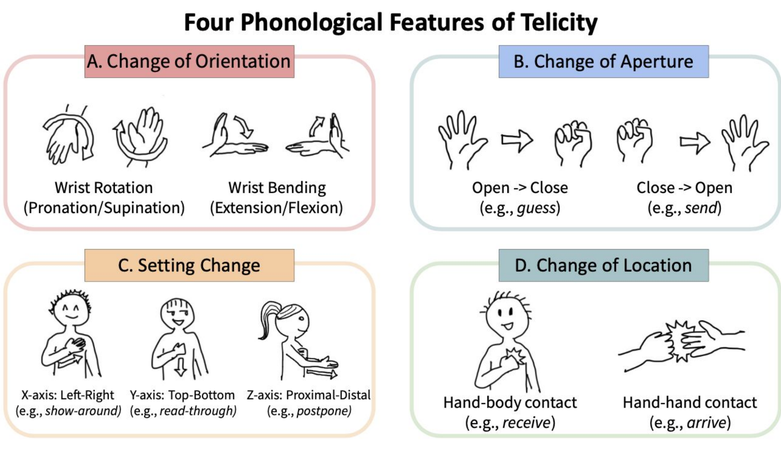
Background

Telicity: whether an event is bounded in time

- Non-signing children and adults can distinguish between telic and atelic signs in ASL (American Sign Language)
- What features enable them to do this?

Prior work has suggested:

- **Four** broad categories of ASL **telic phonological** features
- Kinematic **motion information** can mark telicity

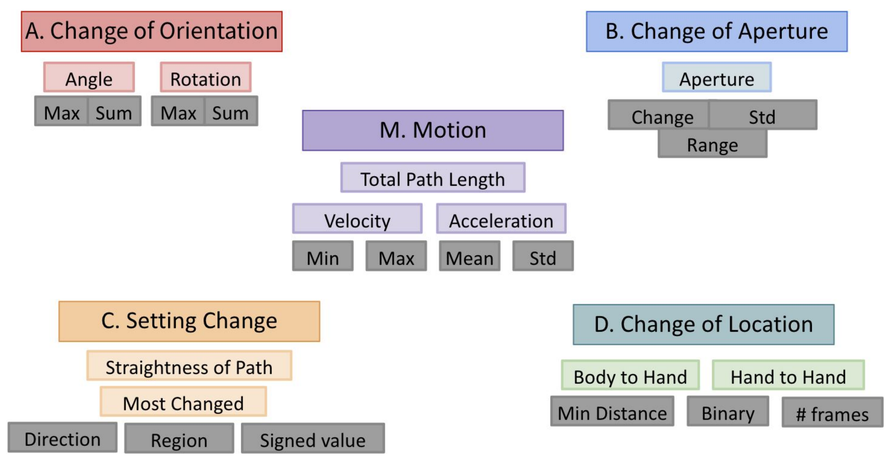


Data and Preprocessing

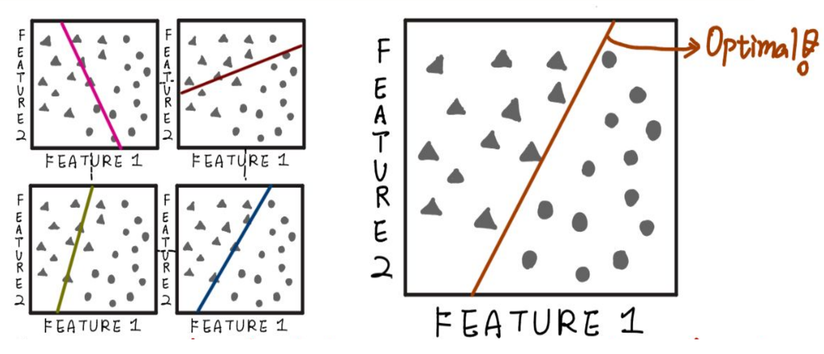
A corpus of verb sign videos: 50 telic, 50 atelic.

- Each sign is a single verb (phrase).
- Use Mediapipe to extract 25 metrics for each sign

Feature Operationalization



Methods: LSH & SVM



LSH: Can we directly cluster signs by telicity with random planes?

SVM: Can we learn how to optimally linearly separate signs with features?

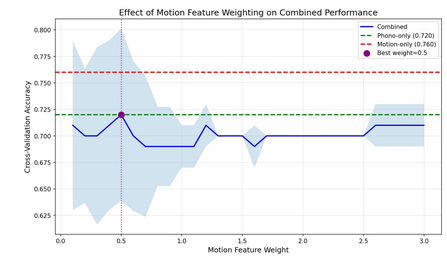
Locality Sensitive Hashing (LSH) Results

	Precision	Recall	F1
Only Motion	0.50	0.88	0.639
Only Phonological	0.51	0.76	0.611
Motion + Phono	0.50	0.88	0.638

- Low precisions indicate that signs of different telicity often have similar feature values

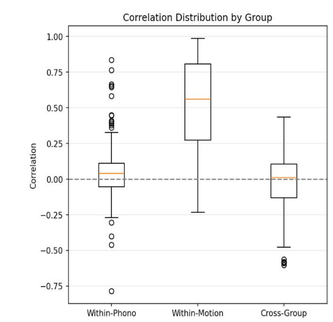
Support Vector Machine (SVM) Results

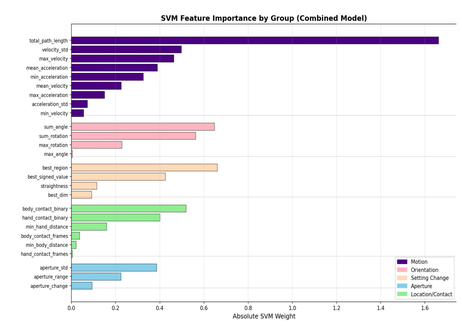
Above 70% accuracy on five-fold cross-validation.



- Using only Motion features produces higher accuracy than only phonological features or combined features

- Motion features are correlated with one another and that there are no strong correlations along other features





- Feature Contribution

Motion > Change of Orientation > Setting Change > Change of Location > Change of Aperture

Three metrics of Change of Location feature are unhelpful

Telic signs have lower sum of angles/rotation than atelic

Conclusions

Motion features are interchangeable, but not phonological

Motion feature > Phonological features

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